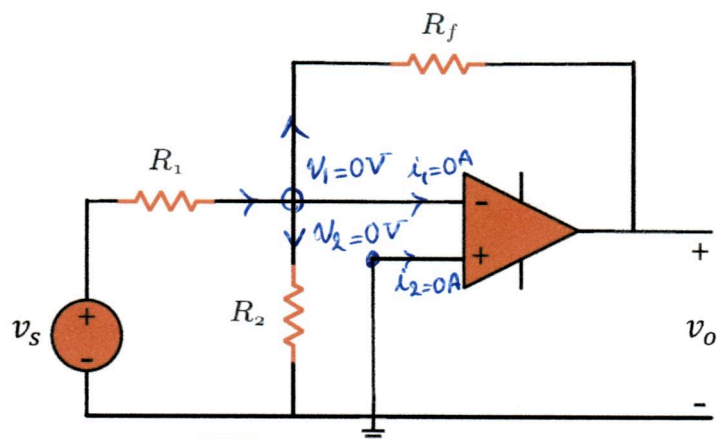


Topic 6: Operational Amplifiers

1. Find the voltage gain $\frac{v_o}{v_s}$ in the following operational amplifier circuit.

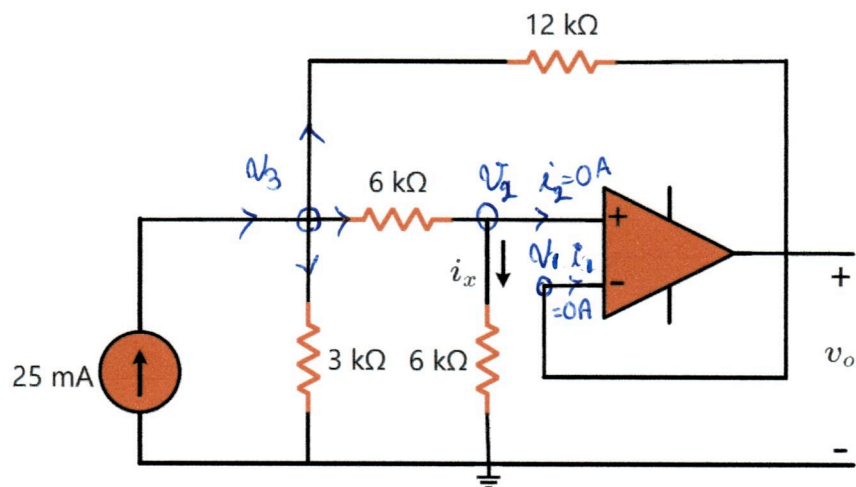


Ideal Op Amp $\rightarrow i_1 = i_2 = 0A$
 v_2 is grounded $\rightarrow v_1 = v_2 = 0V$

Answer: $\frac{v_o}{v_s} = -\frac{R_f}{R_1}$

Solution: KCL @ v_1 : $\frac{v_s - v_1}{R_1} = \frac{v_1}{R_2} + \frac{v_1 - v_o}{R_f}$ $\xrightarrow{v_1=0}$ $\frac{v_s}{R_1} = -\frac{v_o}{R_f} \Rightarrow \frac{v_o}{v_s} = -\frac{R_f}{R_1}$
 Another form of inverting Op Amp.

2. Find the current i_x in the following operational amplifier circuit.



Answer: $i_x = 4.545 \text{ mA}$,

Solution: Ideal opAMP $\rightarrow i_1 = i_2 = 0A$, $V_1 = V_0 \Rightarrow V_2 = V_1 = V_0$

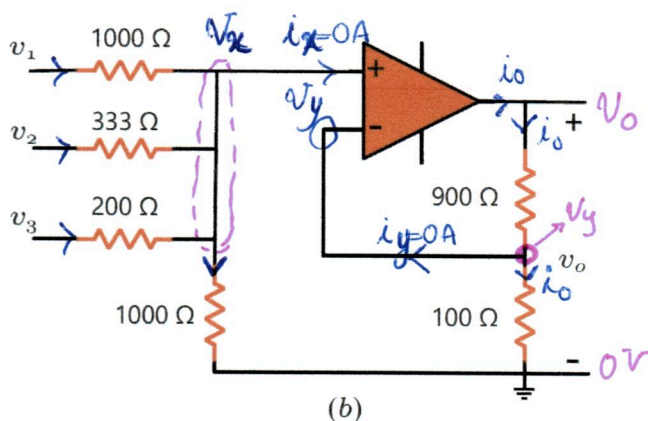
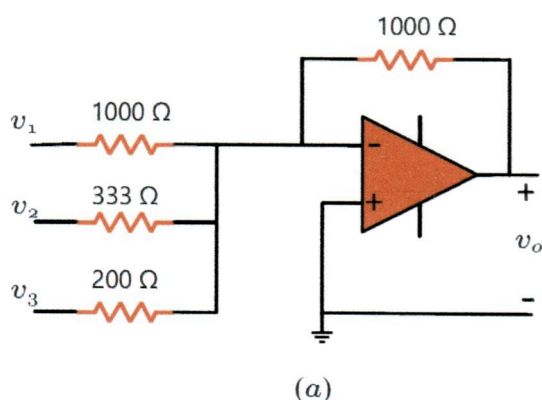
KCL @ V_2 : $\frac{V_2}{6K} = \frac{V_3 - V_2}{6K} \xrightarrow{V_2 = V_0} V_0 = V_3 - V_0 \rightarrow V_3 = 2V_0$

KCL @ V_3 : $25mA = \frac{V_3}{3K} + \frac{V_3 - V_2}{6K} + \frac{V_3 - V_0}{12K} \xrightarrow{V_2 = V_0} 300 = 4V_3 + 2V_3 - 2V_0 + V_3 - V_0$
 $\rightarrow 7V_3 - 3V_0 = 300$

$\Rightarrow \begin{cases} V_3 = 2V_0 \\ 7V_3 - 3V_0 = 300 \end{cases} \Rightarrow 14V_0 - 3V_0 = 300 \Rightarrow V_0 = \frac{300}{11} = 27.27V$

$\Rightarrow i_x = \frac{V_2}{6K} = \frac{V_0}{6K} = \frac{27.27}{6K} = 4.54mA$

3. Calculate v_o in the following two Op Amp circuits in terms of three input voltages v_1 , v_2 , and v_3 .



Answer:

a) $v_o = -(v_1 + 3v_2 + 5v_3)$

b) $v_o = v_1 + 3v_2 + 5v_3$

Solution: a) Typical Summing Op Amp $\rightarrow V_o = -\left(\frac{1K}{1K}v_1 + \frac{1K}{0.333K}v_2 + \frac{1K}{0.2K}v_3\right)$
 $\Rightarrow V_o = -(v_1 + 3v_2 + 5v_3)$ (Note: $1000/333 \approx 3$)
 Ideal op Amp $\rightarrow i_1 = i_2 = 0$, $V_x = V_y$ (I)

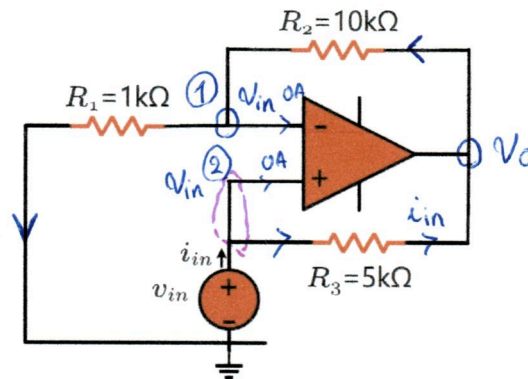
b) KCL @ V_x : $\frac{V_1 - V_x}{1000} + \frac{V_2 - V_x}{333} + \frac{V_3 - V_x}{200} = \frac{V_x}{1000} \xrightarrow{\times 1000} V_1 - V_x + 3V_2 - 3V_x + 5V_3 - 5V_x = V_x$
 $\Rightarrow V_x = \frac{V_1 + 3V_2 + 5V_3}{10}$ (II)

KCL @ V_y : $\frac{V_o - V_y}{900} = \frac{V_y}{100} \xrightarrow{\times 900} V_o - V_y = 9V_y \rightarrow V_o = 10V_y$ (III)
 or $(i_o = \frac{V_o}{100+900} = \frac{V_y}{100} \rightarrow \frac{V_o}{1000} = \frac{V_y}{100} \xrightarrow{\times 1000} V_o = 10V_y)$

(I) $V_x = V_y$ (II) (III) $\rightarrow V_x = \frac{V_o}{10} \rightarrow \frac{V_o}{10} = \frac{V_1 + 3V_2 + 5V_3}{10} \rightarrow V_o = V_1 + 3V_2 + 5V_3$

4. In the following circuit,

- a) Calculate the ratio $\frac{v_{in}}{i_{in}}$ in terms of R_1 , R_2 , and R_3 , and then use the numerical values of resistances given in the circuit to determine the ratio. What does this ratio represent?
- b) If $R_1 = R_2 = R$, determine the value of R_3 for the Op Amp circuit such that $\frac{v_{in}}{i_{in}} = -33 \text{ k}\Omega$



Answer:

a) $\frac{v_{in}}{i_{in}} = -\frac{R_1 R_3}{R_2}$

b) $R_3 = 33 \text{ k}\Omega$

Solution: Ideal op amp $V^- = V^+ = V_{in}$ $i^- = i^+ = 0 \text{ A}$

a) KCL @ node ①: $\frac{V_o - V_{in}}{R_2} = \frac{V_{in} - 0}{R_1} \xrightarrow{\times R_2} V_o - V_{in} = \frac{R_2}{R_1} V_{in} \rightarrow V_o = \left(1 + \frac{R_2}{R_1}\right) V_{in}$ (I)

KCL @ node ②: $i_{in} = \frac{V_{in} - V_o}{R_3} \xrightarrow{\text{(I)}} i_{in} = \frac{V_{in}}{R_3} - \frac{1}{R_3} \left(1 + \frac{R_2}{R_1}\right) V_{in}$

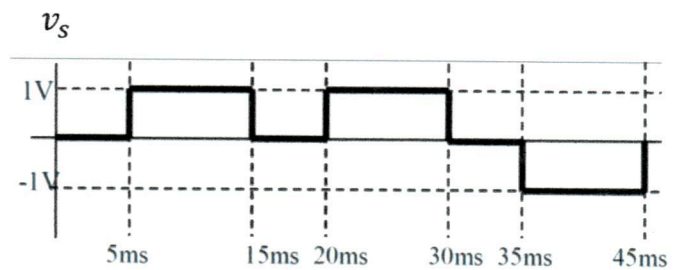
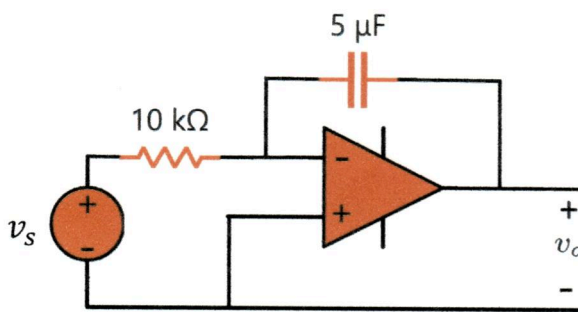
$\rightarrow i_{in} = \frac{V_{in}}{R_3} \left[1 - 1 - \frac{R_2}{R_1}\right] = -\frac{V_{in}}{R_3} \frac{R_2}{R_1} \Rightarrow \frac{V_{in}}{i_{in}} = -\frac{R_1 R_3}{R_2}$

$R_1 = 1 \text{ k}\Omega, R_2 = 10 \text{ k}\Omega, R_3 = 5 \text{ k}\Omega \rightarrow \frac{V_{in}}{i_{in}} = -\frac{1 \text{ k} \times 5 \text{ k}}{10 \text{ k}} = -\frac{5 \times 10^6}{10 \times 10^3} = -500 \Omega$

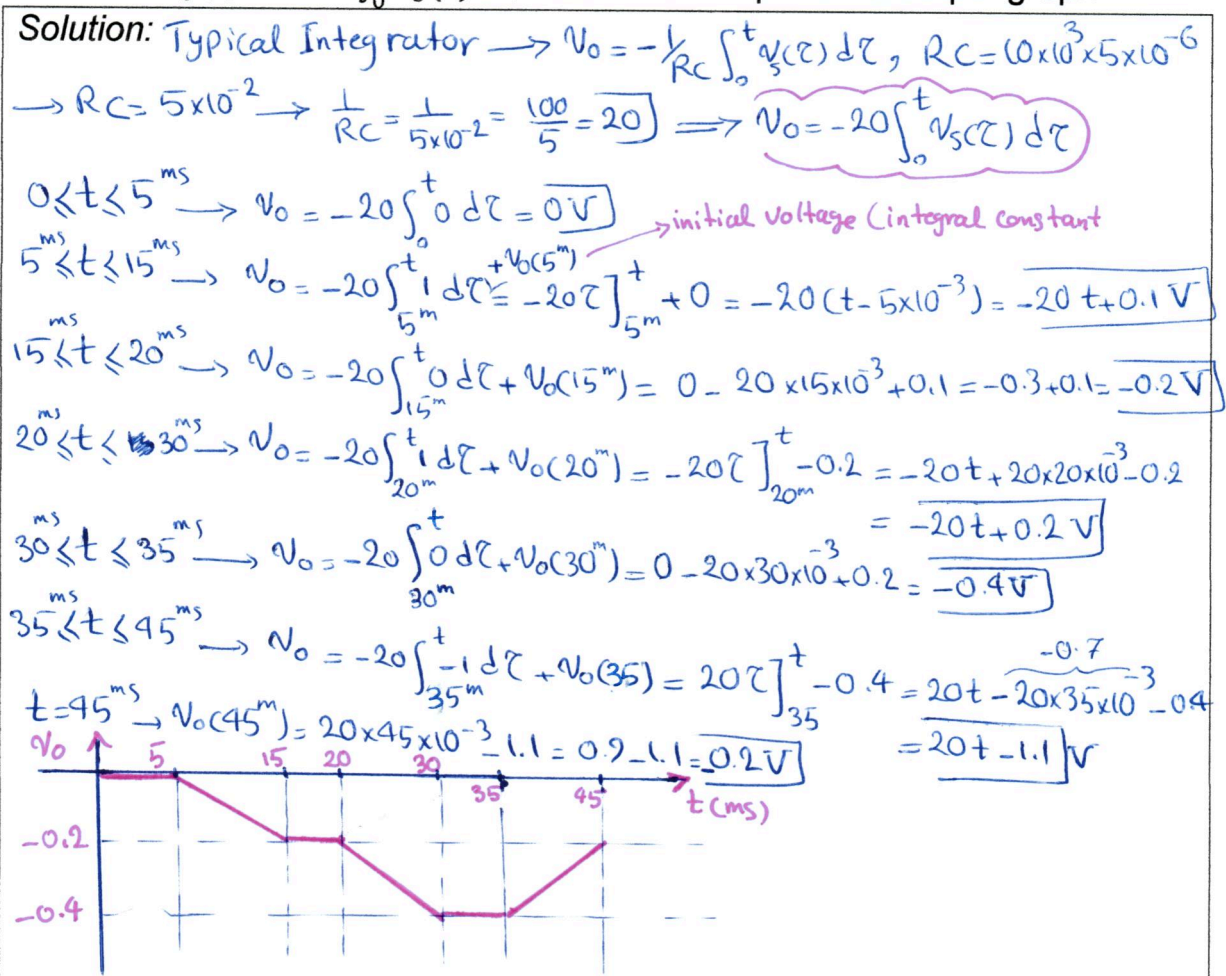
According to Ohm's Law: $V = Ri \rightarrow R = \frac{V}{i}$, Thus $\frac{V_{in}}{i_{in}} = -500 \Omega$ is a negative resistance circuit.

b) if $R_1 = R_2 = R \rightarrow \frac{V_{in}}{i_{in}} = -\frac{R R_3}{R} = -R_3 \Rightarrow \text{choose } R_3 = 33 \text{ k}\Omega$

5. Calculate and draw the output voltage v_o in the following circuit for input voltage signal v_i given in the graph from $0 \leq t \leq 45$ ms.

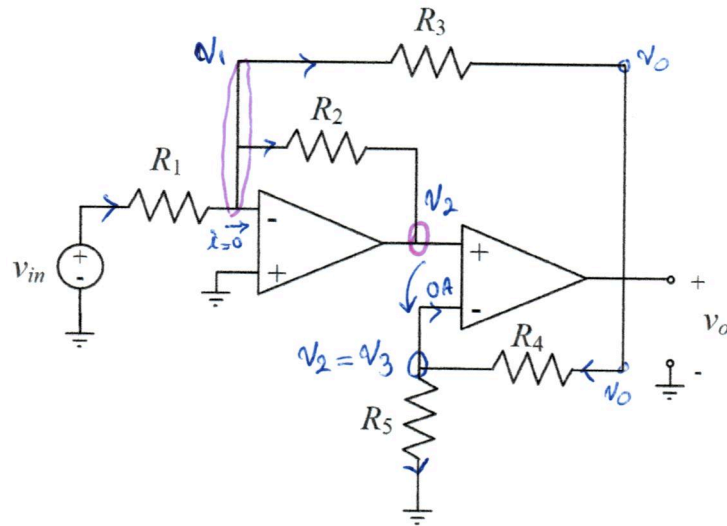


Answer: $v_o(t) = -20 \int_0^t v_s(\tau) d\tau$, You must complete the output graph



6. (Final Exam – S1, 2014) Consider the Op Amp circuit below with input v_{in} and output v_o .

- Derive an expression for the voltage gain $\frac{v_o}{v_{in}}$ in terms of the resistor values R_1, R_2, R_3, R_4 , and R_5 .
- If $R_1 = 1$ k Ω , $R_2 = 2$ k Ω , $R_3 = 3$ k Ω , and $R_5 = 4$ k Ω , determine the value of R_4 such that the voltage gain $\frac{v_o}{v_{in}} = -1.8$.



Answer:

a) $\frac{v_o}{v_{in}} = -\frac{R_2 R_3 (R_4 + R_5)}{R_1 R_3 R_5 + R_1 R_2 (R_4 + R_5)}$

b) $R_4 = 5 \text{ k}\Omega$

Solution:

a) KCL @ v_1 : $\frac{v_{in} - v_1}{R_1} = \frac{v_1 - v_2}{R_2} + \frac{v_1 - v_o}{R_3}$ Ideal op Amp $\frac{v_{in}}{R_1} = -\frac{v_2}{R_2} - \frac{v_o}{R_3}$ $v_1 = 0V$ (I)

KCL @ v_3 : $\frac{v_o - v_3}{R_4} = \frac{v_3}{R_5}$ $\frac{v_3 = v_2}{\times R_4} \rightarrow v_o - v_2 = \frac{R_4}{R_5} v_2 \rightarrow v_o = (1 + \frac{R_4}{R_5}) v_2$ (II)

(II) $\rightarrow v_2 = \frac{R_5}{R_4 + R_5} v_o$ (III)

Non-inverting Op Amp
From v_2 to v_o

Sub (III) to (I) $\frac{v_{in}}{R_1} = -\frac{1}{R_2} \left(\frac{R_5}{R_4 + R_5} \right) v_o - \frac{v_o}{R_3} =$

$\times R_1 \rightarrow v_{in} = -\left(\frac{R_1 R_5}{R_2 (R_4 + R_5)} + \frac{R_1}{R_3} \right) v_o = -\frac{R_1 R_3 R_5 + R_1 R_2 (R_4 + R_5)}{R_2 R_3 (R_4 + R_5)} v_o$

$\Rightarrow \frac{v_o}{v_{in}} = -\frac{R_2 R_3 (R_4 + R_5)}{R_1 R_3 R_5 + R_1 R_2 (R_4 + R_5)}$

All resistors in $k\Omega$
Factor out $k\Omega$ s

b) $\frac{v_o}{v_{in}} = -1.8 = -\frac{2k3k(R_4 + 4k)}{1k \times 3k \times 4k + 1k \times 2k(R_4 + 4k)} = \frac{6(R_4 + 4) \times 10^9}{[12 + 2(R_4 + 4)] \times 10^9}$

$\Rightarrow 1.8 = \frac{6R_4 + 24}{12 + 2R_4 + 8} \rightarrow 36 + 3.6R_4 = 6R_4 + 24 \rightarrow 12 = 2.4R_4$

$\rightarrow R_4 = \frac{12}{2.4} = 5 \text{ k}\Omega$

7. Design a circuit with operational amplifiers that can generate the following output

$$v_o = 9v_1 - 6v_2$$

considering that the range of inputs are $1\text{ V} \leq v_1 \leq 2\text{ V}$ and $2\text{ V} \leq v_2 \leq 3\text{ V}$. The operational amplifiers have a supply voltage of $\pm 12\text{ V}$.

Hint: Consider the impact that the supply voltage of the operational amplifier has on the output of the device and how you can design around this limitation. This means that you have to break the design in two steps:

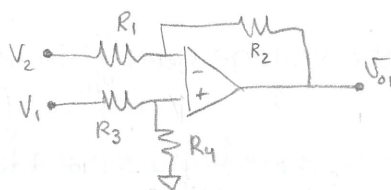
Step1: Design an Op Amp circuit for $v_{o1} = 3v_1 - 2v_2$ as first Op amp stage,

Step2: Cascade the Op Amp circuit in Step 1 to another Op Amp stage for $v_o = 3v_{o1}$.

* STEP 1 : $V_{o1} = 3V_1 - 2V_2$

Different options:

a) Using difference amplifier



From lecture slides (or doing nodal analysis):

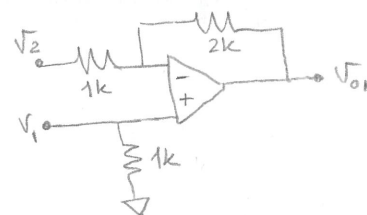
$$V_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} V_1 - \frac{R_2}{R_1} V_2$$

We want $V_{o1} = 3V_1 - 2V_2$, so:

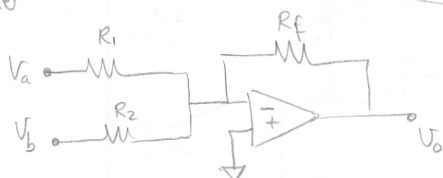
$$\frac{R_2}{R_1} = 2 \Rightarrow \begin{matrix} R_1 = 1\text{k} \\ R_2 = 2\text{k} \end{matrix}$$

$$\frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} = 3 \Rightarrow \frac{2(1 + 1/2)}{1 + R_3/R_4} = 3 \Rightarrow \frac{3}{1 + R_3/R_4} = 3 \Rightarrow \frac{R_3}{R_4} = 0$$

$$\downarrow \\ R_3 = 0 \\ R_4 = 1\text{k}$$



b) Using summing amplifier



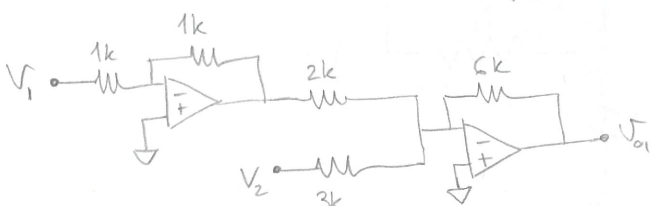
$$V_o = -\frac{R_f}{R_1} V_a - \frac{R_f}{R_2} V_b$$

If we want $V_{o1} = 3V_1 - 2V_2$

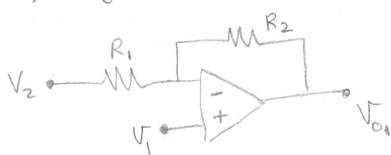
$V_a = -V_1 \Rightarrow$ we need an inverting amplifier with gain 1 first.

$$V_b = V_2$$

$$\left. \begin{matrix} R_f/R_1 = 3 \\ R_f/R_2 = 2 \end{matrix} \right\} \Rightarrow \begin{matrix} \text{If we choose } R_f = 6\text{k}\Omega \\ \text{Then, } R_1 = 2\text{k}\Omega \\ R_2 = 3\text{k}\Omega \end{matrix}$$



c) Using superposition



$$V_{o1} = -\frac{R_2}{R_1} V_2 + \left(1 + \frac{R_2}{R_1}\right) V_1 = -2V_2 + 3V_1$$

$V_1 = 0 \Rightarrow$ inverting amplifier

$V_2 = 0 \Rightarrow$ non-inverting amplifier

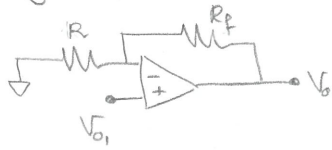
$\hookrightarrow$ We get 2 equations

$$\left. \begin{matrix} -\frac{R_2}{R_1} = -2 \\ 1 + \frac{R_2}{R_1} = 3 \end{matrix} \right\} \Rightarrow \begin{matrix} R_1 = 1\text{k} \\ R_2 = 2\text{k} \end{matrix}$$

Note: Options a) and c) might not work for other values of gains (i.e. the simultaneous equations might not have a solution)

* STEP 2 : $V_o = 3V_{o1} = 3(3V_1 - 2V_2) = 9V_1 - 6V_2$

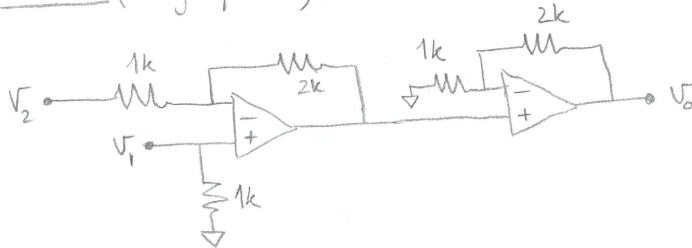
Non-inverting amplifier with gain 3



$$V_o = (1 + R_f/R) V_{o1} = 3V_{o1}$$

$$1 + R_f/R = 3 \Rightarrow R_f = 2k, R = 1k$$

* FULL DESIGN (using option 2)



Note: Using Op Amp circuits in a cascade help to make the circuit more robust,

In any case, a design without cascading would also work for the given range of input voltages:

$$\begin{cases} 1V \leq V_1 \leq 2V \\ 2V \leq V_2 \leq 3V \end{cases} \xrightarrow[\text{(that would lead to highest } V_o)]{\text{worst case scenario}} \begin{matrix} V_1 = 1V \\ V_2 = 3V \end{matrix} \Rightarrow V_o = 9 - 18 = -9 < -12$$

we do not reach saturation

$$\begin{cases} V_{cc}^+ = 12V \\ V_{cc}^- = -12V \end{cases}$$